

Summary - Heterogeneous Multi Agent

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1. Overall Summary

The paper addresses the challenge of optimizing cache storage and content fetching in Cognitive Radio Networks (CRNs) due to unpredictable content popularity and dynamic primary user spectrum occupation. Existing Deep Reinforcement Learning (DRL) approaches often treat the system as a single agent or homogeneous agents, failing to account for the distinct, often conflicting, goals of users (minimizing delivery delay) and cache servers (maximizing revenue/cache efficiency). To solve this, the authors propose a novel heterogeneous Multi-Agent Deep Deterministic Policy Gradient (MADDPG) approach. This model treats users and Secondary Base Stations (SBSs) as two distinct types of agents, enabling them to learn cooperative and competitive strategies simultaneously. The MADDPG framework allows SBSs to optimize the **fraction** of cached content in a continuous action space, enhancing storage efficiency and flexibility compared to discrete action methods. Through centralized training and decentralized execution, agents can optimize actions with partial observations, reducing communication overhead. Numerical simulations demonstrate that the proposed heterogeneous MADDPG significantly reduces user delivery delay and improves SBS cache efficiency, achieving up to 40% higher total reward compared to homogeneous MADRL and outperforming other DRL algorithms like DDPG, PPO, and TD3.

2. Main Idea

The central idea is to apply a heterogeneous Multi-Agent Deep Deterministic Policy Gradient (MADDPG) approach to optimize cooperative caching in Cognitive Radio Networks (CRNs). This involves modeling users and Secondary Base Stations (SBSs) as distinct types of agents with different objectives (users minimize delay, SBSs maximize cache efficiency/revenue) that learn to cooperate and compete simultaneously. The approach allows for continuous optimization of cached content fractions and leverages centralized training with decentralized execution to achieve better system performance in terms of reduced delivery delay and enhanced cache efficiency.

3. Problem Being Solved

Previous DRL approaches for cache optimization in CRNs either model the entire system as a single global agent or treat all cache nodes as homogeneous agents. This overlooks the fact that users and cache servers are autonomous agents with different, potentially conflicting, objectives: users aim to minimize content delivery delay, while cache nodes seek to maximize their reputation or revenue by storing popular content. Additionally, existing methods often struggle with the dynamic and unpredictable nature of content popularity and primary user spectrum occupation, making it difficult to optimize cache storage and content fetching strategies dynamically. The paper also addresses the limitation of discrete action spaces in some DRL methods, which restrict cache management flexibility.

4. Assumptions

- File fragments can be retrieved from a single SBS or the PBS, rather than requiring multiple SBSs to cooperatively relay fragments.
- Secondary Base Stations (SBSs) are geographically closer to users than the Primary Base Station (PBS).
- The transmission rate of SBSs is faster than that of the PBS ($r_{n,k} > r_{o,k}$).
- The time slot duration (T_o) is long enough to serve all content requests within a single time slot.
- Contents are modeled as static, but the number of requests ($q_{k,f}$) is time-variant with respect to users.
- Files can be divided into fragments using rateless codes (e.g., Raptor codes).

5. Limitations

- The complexity of implementing and managing a multi-agent system with heterogeneous agents in a real-world CRN, especially regarding the centralized training phase requiring global observation and action information.
- The assumption of static content popularity for modeling, even though request numbers are time-variant, might simplify the real-world dynamics.
- The specific simulation parameters (e.g., number of SBSs and users, network topology, file distribution) might not generalize perfectly to all CRN scenarios.
- The performance gains are demonstrated in a simulated environment, and real-world deployment might face additional challenges not captured in the simulation.

6. Results and Conclusion

The numerical simulations demonstrate that the proposed heterogeneous MADDPG algorithm effectively reduces user delivery delay and enhances the cache efficiency of SBSs. The algorithm successfully converges for both user and SBS rewards over training episodes. Key findings include: the approach is effective, with SBSs adapting their caching strategy based on geographic location and content popularity; farther SBSs prioritize hotter content, and cached fraction is directly proportional to content popularity. Increasing the replacement cost (β) shifts the popularity-fraction curves from up-concave to down-convex, indicating that SBSs prioritize caching more popular content when cache space is limited. The heterogeneous MADDPG significantly outperforms homogeneous MADDPG (where only SBSs learn or only users learn) by improving the total reward (sum of user and SBS rewards) by nearly 40%. This is attributed to its ability to balance the mixed cooperative-competitive relationship among agents. Finally, the proposed MADDPG consistently achieves higher global rewards compared to other actor-critic based off-policy DRL algorithms, including DDPG, PPO, and TD3, demonstrating its superior performance in optimizing overall system reward by simultaneously optimizing content storage and delivery delay.

7. Real World Impact

This research has practical applications in enhancing the efficiency and performance of next-generation wireless networks, particularly in scenarios involving edge computing and dynamic spectrum access. It can improve content delivery and caching at the network edge, crucial for applications requiring low latency and high bandwidth. This research can also optimize content distribution for various IoT devices and smart city services, reducing network congestion and improving responsiveness. Furthermore, it enhances user experience in mobile networks by reducing content delivery delay and improving network resource utilization through intelligent caching, and facilitates more efficient use of spectrum resources by enabling secondary networks to opportunistically cache and serve content, reducing reliance on the primary network during peak times.

8. Graphs and Figures Analysis

Figure 1: System model

This figure illustrates the overall architecture of the cognitive radio network system. It shows a central Primary Base Station (PBS) connected to multiple Secondary Base Stations (SBSs), which in turn serve various Users. The diagram highlights two types of access: direct access from PBS (secondary) and preferred access from SBSs (firstly). Crucially, it depicts the multi-agent learning framework, showing K agents for users and N agents for SBSs. Each agent type has an 'Actor' network (π) for action selection and a 'Critic' network (Q) for evaluating actions, with separate reward (R) and state (S) inputs. The 'Training' phase involves both Actor and Critic, while 'Execution' primarily uses the Actor. This model underpins the heterogeneous MADDPG approach.

Figure 2: Locations and rewards of heterogeneous agents

This figure is composed of three sub-figures illustrating the simulation setup and the convergence of rewards. (a) Position Distribution: This scatter plot shows the spatial arrangement of the network components. The PBS is located at (0,0). Five SBSs (SBS1-SBS5) and three Users (User1-User3) are distributed around the PBS. This visualizes the "nearby" assumption for SBSs relative to users and their distance from the PBS. (b) Users' reward: This line graph plots the average reward for each of the three users (User 1, User 2, User 3) against the number of training episodes. All user rewards show a clear upward trend and converge to a stable, higher value after approximately 200,000 episodes, indicating that the learning algorithm successfully optimizes user-specific objectives (minimizing delay). (c) SBSs' reward: This line graph plots the average reward for each of the five SBSs (SBS 1-SBS 5) against the number of training episodes. Similar to user rewards, all SBS rewards converge to stable, positive values, demonstrating that the algorithm effectively optimizes SBS-specific objectives (maximizing cache efficiency and revenue). SBS2 and SBS4 show slightly lower rewards, which is later explained by their geographic location.

Figure 3: Contents popularity vs. cached fraction

This figure consists of two sub-figures analyzing the relationship between content popularity and cached fraction under different conditions. (a) Impact by location of SBS: This line graph shows the average cached fraction (λ) as a function of normalized content popularity for each of the five SBSs. It reveals that SBSs farther from users (SBS2 and SBS4, as indicated in the text) exhibit different popularity-fraction curves and generally lower rewards. These SBSs tend to cache a larger fraction of the most popular contents, sacrificing less popular ones, to compensate for their less favorable geographic position and attract more users. The trend for all SBSs is that more popular content is stored in larger fractions. (b) Impact by factor β : This line graph shows the average cached fraction (λ) as a function of normalized content popularity for different values of the replacement cost factor ($\beta = 0.5, 0.8, 1.0, 1.2$). As β increases, the curves transition from an up-concave shape to a down-convex shape. This indicates that with higher replacement costs, SBSs prioritize caching more popular content, especially when cache space is scarce, aligning with the model's expectation of maximizing storage efficiency for hot content.

Figure 4: Comparison with homogeneous MADDPG

This figure presents two line graphs comparing the overall rewards of users and SBSs under different learning scenarios, including the proposed heterogeneous MADDPG and two homogeneous MADDPG baselines. (a) Users' overall reward: This graph plots the sum of user rewards against training episodes. The "Heter-MADDPG (users and SBSs)" (blue line) shows the highest and most stable converged reward. The "Homo-MADDPG (users Only)" (green line) where only users learn, performs worse. The "Homo-MADDPG (SBSs Only)" (red line) where only SBSs learn, performs the worst for users. This demonstrates the benefit of heterogeneous learning for user satisfaction. (b) SBSs' overall reward: This graph plots the sum of SBS rewards against training episodes. Similarly, the "Heter-MADDPG (users and SBSs)" (blue line) achieves the highest and most stable converged reward. The "Homo-MADDPG (SBSs Only)" (red line) performs better than "Homo-MADDPG (users Only)" (green line) for SBSs, but still significantly worse than the heterogeneous approach. This highlights that considering both agent types simultaneously leads to better outcomes for SBSs.

Figure 5: Performance comparison with other DRL

This line graph compares the global reward (sum of all user and SBS rewards) of the proposed heterogeneous MADDPG against several other state-of-the-art DRL algorithms: DDPG, PPO, and TD3. The x-axis represents training episodes, and the y-axis represents the global reward. The "MADDPG" (blue line) consistently achieves the highest global reward and converges faster and to a higher value than DDPG, PPO, and TD3. This figure provides strong evidence for the superior performance of the heterogeneous MADDPG approach in optimizing the overall system reward in a mixed cooperative-competitive environment.